

An action floor on transverse phase-space area produces the Yang–Mills quantum-mechanical gap, and a gap forces an action unit

In the Yang–Mills quantum-mechanical model of G07 the two directions of the analogy between a positive action floor $h > 0$ and a mass gap can both be stated and proved inside one solved system. Forward: classical mechanics plus the single postulate that no transverse oscillation has phase-space area below $h/2$ confines the escaping valley motion to $|x| \leq 2\sqrt{m} E/(\hbar g)$ at energy E , makes the accessible phase-space volume finite with the growth $E^{3/2} \ln E$ that Simon records for the true eigenvalue counting function, and reproduces through Bohr–Sommerfeld quantization the energy unit $\varepsilon = \hbar^{4/3} g^{2/3} m^{-2/3}$ of the exact spectrum; the operator inequality behind G07’s Theorem 5 is this postulate made rigorous by the uncertainty inequality. The abelian model has no transverse oscillation, so the same floor constrains nothing and its spectrum stays continuous: with $h > 0$ assumed, a commutative field still has no gap. Backward: in the classical model with constants m and g , any positive finite gap Δ , whatever extra constant produces it, supplies the action unit $\Delta^{3/4} m^{1/2} g^{-1/2}$, so within the model “gap” and “action unit” are dimensionally equivalent. The same equivalence holds for Yang–Mills in $d = 2$ and $d = 3$, where the classical coupling is dimensionful, and fails exactly in $d = 4$, the action-critical dimension, where a mass gap together with c admits no action unit. The floor postulate is a phase-space area statement; the necessity theorem sought in STATE item 1 is the derivation of such a floor. This is G08, exploratory, with no ledger promotion.

1. The transverse phase-space area and the floor postulate

Take the scalar model of G07, Theorem 5, with classical Hamiltonian

$$H_{\text{cl}} = \frac{p_x^2 + p_y^2}{2m} + \frac{g^2}{2} x^2 y^2.$$

At fixed $x \neq 0$ the transverse motion is the oscillator $E_y = p_y^2/2m + \frac{1}{2}m\omega_x^2 y^2$ with $\omega_x = g|x|/\sqrt{m}$.

Proposition 1. The transverse orbit at fixed x and transverse energy E_y is the ellipse $\{p_y^2/2m + \frac{1}{2}m\omega_x^2 y^2 = E_y\}$, and its enclosed phase-space area is

$$J_y = \oint p_y dy = \frac{2\pi E_y}{\omega_x} = \frac{2\pi\sqrt{m} E_y}{g|x|}.$$

Proof. The ellipse has semi-axes $\sqrt{2mE_y}$ in p_y and $\sqrt{2E_y/(m\omega_x^2)}$ in y ; the product times π gives $2\pi E_y/\omega_x$. $\square$

J_y is the adiabatic invariant of the transverse oscillation: along the valley it is conserved to leading order when ω_x changes slowly on the oscillation’s time scale, $|\dot{x}/x| \ll \omega_x$, that is $|x| \gg x_{\text{ad}}(E) = (2E)^{1/4} g^{-1/2}$ at energy E (Landau–Lifshitz, *Mechanics* §49; the note uses only Proposition 1 and the inequality below, which need no adiabatic limit).

Floor postulate $F(h)$. No admitted state has a transverse oscillation of phase-space area below $h/2 = \pi\hbar$: $J_y \geq h/2$ whenever $x \neq 0$, and $J_x \geq h/2$ whenever $y \neq 0$.

The value $h/2$ is the Bohr–Sommerfeld ground-state area $(n + \frac{1}{2})h$ at $n = 0$; any positive multiple of h gives the same conclusions with a different pure number.

2. Forward: the floor confines the valley and produces the gap

Theorem 2. Under $F(h)$, every admitted classical state of energy E satisfies

$$|x| \leq X(E) = \frac{2\sqrt{m} E}{\hbar g}, \quad |y| \leq X(E),$$

and the transverse energy obeys $E_y \geq \hbar\omega_x/2 = \hbar g|x|/(2\sqrt{m})$. In particular the admitted region of the energy shell is bounded, whereas without the floor the shell contains the unbounded valley motions $y = p_y = 0$, $x(t) = x_0 + p_x t/m$ of every energy $p_x^2/2m > 0$.

Proof. For $x \neq 0$, Proposition 1 and $E_y \leq E$ give $J_y = 2\pi\sqrt{m} E_y/(g|x|) \leq 2\pi\sqrt{m} E/(g|x|)$; the floor $J_y \geq \pi\hbar$ then forces $|x| \leq 2\sqrt{m}E/(\hbar g)$. Reading the same inequality as a bound on E_y gives $E_y \geq \hbar g|x|/(2\sqrt{m})$. The y bound is symmetric. $\square$

Remarks.

1. *The floor is the quantum bound.* The inequality $E_y \geq \hbar g|x|/(2\sqrt{m})$ is word for word the oscillator bound of G07, Theorem 5 (Lemma 0 there): $-\frac{\hbar^2}{2m}\partial_y^2 + \frac{g^2}{2}x^2y^2 \geq \hbar g|x|/(2\sqrt{m})$. The quantum theorem is the floor postulate enforced by the uncertainty inequality $\int(\mu|\phi'|^2 + \nu y^2|\phi|^2) \geq \sqrt{\mu\nu} \int|\phi|^2$, which is the statement that no normalizable state occupies a transverse phase-space area below $\hbar/2$. So in this model the implication “ $\hbar > 0$ gives the gap” is exact, with $\hbar$ entering only through the area floor, and the rest of quantum kinematics is not used.
2. *Self-consistency of the adiabatic reading.* The cutoff $X(E)$ lies in the adiabatic region: $X(E)/x_{\text{ad}}(E) = 2^{3/4}(E/\varepsilon)^{3/4}$ with $\varepsilon = \hbar^{4/3}g^{2/3}m^{-2/3}$, so for $E \gg \varepsilon$ the floor acts where J_y is conserved, and near the ground state, $E \sim \varepsilon$, the classical reading is only an estimate.
3. *Counting.* Without the floor the phase-space volume $|\{H_{\text{cl}} \leq E\}|$ is infinite for every $E > 0$ (G07, Theorem 5, proof of (iv)). With the floor, the transverse area at fixed x and fixed p_x is $2\pi(E - p_x^2/2m)/\omega_x$; integrating over $|p_x| \leq \sqrt{2mE}$ gives $(8\pi/3)\sqrt{2m} E^{3/2}\sqrt{m}/(g|x|)$ per unit x , and integrating $|x|$ from a fixed inner scale to $X(E)$ gives

$$|\{H_{\text{cl}} \leq E\} \cap F(\hbar)| = \frac{16\sqrt{2}\pi m}{3g} E^{3/2} \ln \frac{X(E)}{x_0} + O(E^{3/2}),$$

which grows like $E^{3/2} \ln E$. Simon states (Ann. Phys. 146 (1983) 209, p. 3, passage read) that the true counting function $N(E)$ of $-\Delta + x^2y^2$ grows like $E^{3/2} \ln E$, while the classical volume is infinite. The floor therefore reproduces the growth law of the exact spectrum; the identification $N(E) \approx |\{\cdot\}|/\hbar^2$ is the Weyl heuristic, stated here as a consistency check and not as a theorem.

4. *Semiclassical levels.* With $J_y = (n + \frac{1}{2})\hbar$ conserved, the valley motion has effective Hamiltonian $p_x^2/2m + \kappa_n|x|$, $\kappa_n = (n + \frac{1}{2})\hbar g/\sqrt{m}$, and Bohr–Sommerfeld quantization $\oint p_x dx = (N + \frac{1}{2})\hbar$ with $\oint p_x dx = \frac{8\sqrt{2m}}{3} E^{3/2}/\kappa_n$ gives

$$E_{n,N}^{\text{BS}} = \varepsilon \left[\frac{3\pi(2n+1)(2N+1)}{16\sqrt{2}} \right]^{2/3}, \quad \varepsilon = \hbar^{4/3}g^{2/3}m^{-2/3}.$$

The unit ε is the exact one of G07, Theorem 5(4), as Theorem 1 of G07 requires; the pure numbers are semiclassical estimates of e_n and are labelled as such. The lowest level and the first spacing are positive: the old quantum theory of this model already has a gap.

Corollary 3 (the commutative model). For the abelian matrix model, and for $D = 1$, the potential vanishes identically: there is no transverse oscillation, $F(\hbar)$ imposes no condition, the classical energy shells stay unbounded and the quantum spectrum is $[0, \infty)$ (G07, Theorem 6(5)). An action floor produces a gap only in the presence of the commutator potential, which is what supplies the transverse ellipses whose area the floor bounds.

For the $\text{SU}(2)$ model with $D \geq 2$ the same argument applies to each of the two transverse directions of $\vec{x}_j$ relative to $\vec{x}_i$: the floor gives $E_\perp \geq \hbar g|\vec{x}_i|/\sqrt{m}$, the bound of G07, Theorem 6(1), and $|\vec{x}_i| \leq \sqrt{m} E/(\hbar g)$ on the energy shell.

3. Backward: a gap forces an action unit, except in the action-critical case

Proposition 4. Let the classical model with fixed constants m, g (dimension vectors $d_m = (1, 0, 0)$, $d_g = (\frac{1}{2}, -1, -1)$) be enlarged by any fixed constants, and suppose the enlarged theory has a positive finite gap Δ depending on the fixed constants only. Then the theory has the action unit

$$A = \Delta^{3/4} m^{1/2} g^{-1/2},$$

and conversely an action unit A gives the energy unit $A^{4/3} g^{2/3} m^{-2/3}$.

Proof. Energy has dimension vector $(1, 2, -2)$ and action $(1, 2, -1)$. The system $a d_m + b d_g + c(1, 2, -2) = (1, 2, -1)$ has the unique solution $a = \frac{1}{2}$, $b = -\frac{1}{2}$, $c = \frac{3}{4}$: the length and time components give $-b + 2c = 2$, $-b - 2c = -1$, hence $c = \frac{3}{4}$, $b = -\frac{1}{2}$, and the mass component gives $a = \frac{1}{2}$. So $m^{1/2} g^{-1/2} \Delta^{3/4}$ has action dimension whatever the extra constants are. The converse is G07, Theorem 3 with $k = 4$. $\square$

Within the model, therefore, “the gap is positive” and “an action unit exists” are the same dimensional statement, and G07’s Theorem 5 supplies the dynamical content, $0 < \delta_1 < \infty$, that turns the unit into the number $\Delta = \delta_1 \varepsilon$.

Proposition 5 (which field theories share this equivalence). With the classical coupling $1/g^2$ of dimension action-length $^{4-d}$, the speed c and a gap Δ , an action unit exists in $d = 2$ and $d = 3$ and does not exist in $d = 4$:

d	fixed classical constants	action unit from $(1/g^2, c, \Delta)$
2	$1/g^2 \sim (1, 4, -1), c$	$(1/g^2)^{1/3} c^{-2/3} \Delta^{2/3}$
3	$1/g^2 \sim (1, 3, -1), c$	$(1/g^2)^{1/2} c^{-1/2} \Delta^{1/2}$
4	c only	none: $a(0, 1, -1) + b(1, 2, -2) = (1, 2, -1)$ has no solution
mechanics, $k = -2$	$m, \lambda \sim (1, 4, -2)$	$\sqrt{m\lambda}$ already classical; Δ adds nothing

Proof. Solve $a d_{1/g^2} + b d_c + c'(1, 2, -2) = (1, 2, -1)$: for $d = 3$, adding the length and time equations gives $2a = 1$, then $c' = \frac{1}{2}$, $b = -\frac{1}{2}$; for $d = 2$, $3a = 1$, $c' = \frac{2}{3}$, $b = -\frac{2}{3}$; for $d = 4$ the mass component forces $b = 1$, the length component then forces $a = 0$, and the time component reads $-2 = -1$. The $k = -2$ row is G07, Section 6. $\square$

The equivalence “gap if and only if action unit” thus holds whenever the classical theory carries a dimensionful coupling, and fails exactly at the action-critical dimension, where the classical theory carries no constant at all and a mass gap with c yields a length only after $\hbar$ is supplied. This is the dimensional reason the four-dimensional problem is one-directional: the gap needs a scale generated by transmutation, and that scale does not by itself return an action unit.

4. What this decides for the positive-action question

- **Direction.** In the solved model the analogy is an equivalence at the level of units and an implication at the level of dynamics: the floor postulate, used only through the transverse area, produces the gap (Theorem 2 and Remark 1), and the gap returns the unit (Proposition 4). The mass-gap problem is the forward direction with $\hbar$ and the gauge group given; the necessity problem of STATE item 1 is the derivation of the floor itself. Inside this model those two problems have the same dimensional content, and the necessity problem is exactly as hard as proving a gap from a consistency premise that does not name $\hbar$.
- **The second ingredient.** Corollary 3 states the commutative case with $h > 0$ assumed: an action floor alone gives no gap; the commutator potential is what creates transverse

oscillations for the floor to act on. Any transported statement “ $h > 0$ gives a gap” must name the structure that plays this role.

- **Areas.** The floor is a bound on a phase-space area, the ellipse of Proposition 1. The programme’s anchor, Galileo’s area $vF\tau^3/(6m)$ between the inertial line and the parabola, is a configuration-space area converted to action by the factor $3F/v$ (N01 §1). The model shows what a successful floor would have to look like on the phase-space side: a lower bound on the area of a closed transverse orbit, invariant under the adiabatic deformation of the orbit. A constant-force motion has no closed transverse orbit, which is one more statement of why the Galileo class is similarity-covariant (G07, Proposition 14).
- **Gate.** For gap-track work the supplied/generated gate of G07 can be applied in phase-space terms: a generated gap needs a family of closed orbits whose area can fall below h along a non-compact direction.

5. Strategic consequence

G08 closes the loop inside the Yang–Mills quantum-mechanical model: floor gives gap, gap gives unit, and the commutative model is inert under the floor. Two dimensional facts are retained for the programme: the equivalence of gap and action unit holds exactly when the classical theory has a dimensionful coupling, and fails at $d = 4$; and the phase-space area of a closed transverse orbit is the object a floor must bound. Neither changes STATE item 1’s target, which remains the derivation of the floor from consistency premises; N02 gains the phase-space form of the floor as the statement to aim at. No ledger claim is promoted; Theorem 2 and Propositions 1, 4, 5 are elementary and the semiclassical remarks are labelled estimates. Sources are in B79.